Merve Tomečková Öztoprak

m.oeztoprak@gmail.com
isotrace.github.io/MerveOztoprak | linkedin.com/in/merve-oeztoprak

Professional Summary

Analytical chemist with 5+ years of hands-on laboratory and field experience specializing in GC-MS, LC-MS (Orbitrap), and IRMS method development and analytical testing for the detection and characterization of organic compounds in diverse environmental matrices. During my PhD at NIOZ I developed and validated LC and GC Orbitrap workflows for isotopologue analysis at natural abundance. Background in molecular ecosystem sciences (omics) and evolutionary biology, with experience in working with plants, microbial cultures and soil organism and field experience from Arctic to tropical environments. Skilled in advanced data analytics (R/Python), setting up reducible R pipelines and writing method SOPs and training students. Strong publication record and international collaborative network in organic geochemistry. Motivated to extend intramolecular isotope methods to new targets in forensics, ecology and geochemistry.

Key Skills

- **Mass Spectrometry:** ESI/APCI-LC-Orbitrap MS, GC-Orbitrap MS, GC-MS, LC-MS, GC-FID-MS, Prep-HPLC, IRMS, XRF, EA (Thermo Fisher Scientific, Agilent, Waters) — instrument operation, maintenance, calibration, and troubleshooting. Eager to expand into ICP-MS
- **Method Development & Validation:** End-to-end design, validation, and optimization of novel analytical workflows for trace-level detection of intramolecular isotopic signatures, quantification, and structural characterization of organic compounds in complex matrices
- **Quality Assurance:** GLP (Good Laboratory Practice) familiarity, ISO-compliant workflows, SOP development, quality control procedures, safety compliance, technical reporting, and meticulous record-keeping in commercial and research laboratory settings
- **Data Analytics:** R and Python pipelines for isotope correction algorithms, predictive modeling, multivariate statistics, and network analysis of large MS datasets
- **Fieldwork & Environmental Sampling:** Ecological sampling campaigns across Arctic (Svalbard), tropical (Hong Kong), lacustrine, and marine environments

Professional Experience

Ph.D. Researcher — Analytical Chemistry & Instrumental Analytics. Jan 2020 – Apr 2026

Royal Netherlands Institute for Sea Research (NIOZ), Netherlands

- **Developed and validated novel LC-Orbitrap MS workflows** for site-specific carbon isotopic analysis of isoprenoid lipids, enabling source differentiation at natural abundance
- **Designed custom bioanalytical data pipelines** (R/Python) for high-throughput LC-MS dataset evaluation, including predictive modeling, multivariate statistics, and hierarchical network analysis
- Independently sourced analytical materials and software solutions through industrial partnerships (Thermo Fisher Scientific), optimizing project costs and workflows
- **Maintained, calibrated, and troubleshot Orbitrap MS instrumentation** across multiple projects, ensuring reliable high-resolution performance and minimal downtime
- **Supervised and trained HBO/PhD students** in LC-MS operation, sample preparation, and data interpretation
- Delivered technical reports and scientific presentations to academic, industrial, and governmental audiences
- Established and documented **quality control procedures and SOPs** for novel analytical workflows, ensuring reproducibility, data integrity, and compliance with institutional standards

Independent Researcher May 2019 – Sep 2019

California Institute of Technology (Caltech), USA

- **Pioneered a GC-Orbitrap MS method** for direct measurement of site-specific carbon isotopic signatures in phytane — the first experimental validation of this analytical approach for isoprenoids

Independent Researcher. Oct 2018 – Apr 2019

The Australian National University, Canberra, Australia

- Conducted high-sensitivity inorganic and organic geochemical analyses (**XRF, EA, IRMS, GC-MS**) using ultra-clean methods for lipid extracts from Precambrian sediment cores, supporting paleoenvironmental reconstructions

Independent Researcher Jan 2018 – Apr 2018

Georg-August University Göttingen, Germany

- Analyzed biomarker signatures in sediment cores from Lake Seeburg using GC-MS, contributing to environmental reconstruction studies
- **Produced an expert report for a governmental planning and development project** (Amt für Agrarstruktur, Göttingen) — demonstrating ability to translate analytical results into actionable information for non-scientific stakeholders

Trainee — Environmental Safety Division (Ecotoxicology) Jun 2017 – Sep 2017

Bayer CropScience, Monheim, Germany

- Executed **GLP-compliant acute and chronic toxicity bioassays** on soil model organisms under regulated laboratory conditions
- Developed and presented an automated image-based evaluation system for toxicity assessments, increasing analytical throughput and data quality

Education

Ph.D. in Organic Geochemistry Thesis submission expected in December 2026

Royal Netherlands Institute for Sea Research (NIOZ) / Utrecht University, Netherlands

Thesis: Fragment-specific isotope ratio analysis of isoprenoid lipids using Orbitrap mass spectrometry

M.Sc. in Evolutionary Biology (Double Degree) Oct 2019

University of Uppsala (Sweden) & Ludwig-Maximilian University Munich (Germany)

B.Sc. in Molecular Ecosystem Sciences Aug 2017

Georg-August University Göttingen, Germany

- Erasmus+ Semester: University Centre in Svalbard (UNIS), Norway — Arctic Biology and Environmental Management
- DAAD RISE Research Internship: Chinese University of Hong Kong — Field surveys on earthworm species composition across Hong Kong ecosystems

Publications

Tomečková Öztoprak, M., van der Meer, M., Hopmans, E., Villanueva, L., Schouten, S. Methodological development and theoretical framework for the isotopic analysis of mass spectral fragments of phytol using Orbitrap Mass Spectrometry. *ACS Omega,* 11(18), 26295-26309.

Tomečková Öztoprak, M., van der Meer, M., Schouten, S., Villanueva, L., Eiler, J. Measuring carbon isotopic signatures of mass spectrometric fragment ions of phytane using ultra high resolution orbitrap mass spectrometry. *Journal of the American Society for Mass Spectrometry (in submission)*

Tomečková Öztoprak, M., von Meijenfeldt, F. A., van der Meer, M., van Koeveringe L., Thiel V., Hopmans, E., Schouten, S., Villanueva, L. Fragment stable isotope analysis (FSIA) using Orbitrap mass spectrometry as a new tool to distinguish biological sources of phytol *(in preparation)*

Neubauer, C., Kantnerová, K., Lamothe, A., Savarino, J., Hilkert, A., Juchelka, D., **Öztoprak, M.**, et al. (2023). Discovering nature's fingerprints: isotope ratio analysis on bioanalytical mass spectrometers. *Journal of the American Society for Mass Spectrometry,* 34(4), 525-537.

Zeman-Kuhnert, S., Heim, C., **Öztoprak, M.**, & Thiel, V. (2023). Reconstructing eutrophication trends of a shallow lake environment using biomarker dynamics and sedimentary sterols. *Organic Geochemistry*, 177, 104555.

Selected Talks & Presentations

- Organizing Chair, Gordon Research Seminar in Organic Geochemistry (Aug 2026)
- Invited Speaker, 'Orbitrap Isotope Applications', TU Munich (Apr 2024)
- Oral presenter, BASIS Conference, Amsterdam (Apr 2024)
- Oral presenter, IMOG Conference, Montpellier (Sep 2023)
- Invited speaker, 'Nature's Fingerprints: New Ways to Measure Stable Isotopes', Bremen (Nov 2021)
- Guest Lecturer, University of Bristol (Dec 2021)

Awards & Scholarships

Best Student Presentation Award, IMOG	09/2021
Best Student Poster Award, AOGC	11/2018
PROMOS Research Scholarship	10/2018 – 04/2019
Erasmus Mundus Category B Scholarship for Master studies	09/2017 – 08/2019
DAAD RISE Worldwide Scholarship	07/2016 – 09/2016
Erasmus+ Scholarship, UNIS Svalbard	01/2015 – 06/2015

Leadership & Community Engagement

- Founding member, Inclusion, Diversity, Equity, and Accessibility (IDEA) Group, NIOZ
- Founding member of the Unlearning Racism in Geo- and Earth Sciences (URGE) pod, NIOZ
- Organizing Chair, Gordon Research Seminar in Organic Geochemistry (2026)
- Active contributor to the international Isotopomics community through workshops and invited talks
- Supervised and mentored HBO and PhD students in analytical workflows